

Separator Pressure Anomaly

Field Manual — Setup and Troubleshooting Reference

Use case definition. A production separator removes gas from liquid and controls pressure and liquid level before downstream processing. Abnormal behavior includes high/low pressure, unstable pressure, incorrect control-valve response, gas or liquid flow changes, level-pressure interaction, instrument disagreement, relief events, slugging, and downstream restrictions. Troubleshooting maintains separation performance and protects the vessel and relief system.

Manual setup and use. Use the vessel design pressure, P&ID, control narrative, relief-device basis, and OEM/site alarm or trip limits. Values are training/reference values.

Safety note. Never bypass a relief device or open a separator/manway/line without approved isolation, depressurization, and permit controls.

Quick-use workflow

Use this sequence for every event: establish a safe operating state; capture the initial numerical readings; compare each value with the stated reference; apply the trigger-based action; correct the identified component or process path; restore operation gradually; and confirm the expected numerical result. If an OEM/site alarm or trip limit differs from a training value, the OEM/site limit governs.

Scenario 1: High separator pressure

Field Condition

pressure 11.5 barg from 7.5–9.0; gas flow 4.2 MMSm³/d; valve 18 %

Problem

High pressure indicates downstream restriction, valve failure, or gas-flow surge.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure above 10 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure 7.5–9.0 barg and valve 35–65 %.

Expected Result

pressure 7.5–9.0 barg and valve 35–65 %

Scenario 2: Low separator pressure

Field Condition

pressure 5.2 barg; gas flow 1.8 MMSm³/d from 3.0–4.0; valve 94 %

Problem

Low pressure with valve nearly open suggests low inflow, leaking valve, or control failure.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure below 6.5 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure 7.5–9.0 barg.

Expected Result

pressure 7.5–9.0 barg

Scenario 3: Pressure fluctuation

Field Condition

pressure cycles 7.0–10.8 barg every 45 s; valve 12–82 %; level 48 %

Problem

Control instability, slugging, or bad pressure input is cycling the valve.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses swing above 2 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when swing below 0.5 barg.

Expected Result

swing below 0.5 barg

Scenario 4: Control valve stuck open

Field Condition

valve feedback 100 %; pressure 5.8 barg; gas flow 1.6 MMSm³/d

Problem

A failed-open valve is over-depressuring the separator.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses feedback above 95 % with low pressure.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when valve 35–65 % and pressure 7.5–9.0.

Expected Result

valve 35–65 % and pressure 7.5–9.0

Scenario 5: Control valve stuck closed

Field Condition

valve feedback 4 %; pressure 10.9 barg; downstream 8.2 barg

Problem

A failed-closed valve is restricting gas outlet.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses feedback below 10 % with pressure above 10 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when valve responds through 35–65 % and pressure normal.

Expected Result

valve responds through 35–65 % and pressure normal

Scenario 6: Downstream gas restriction

Field Condition

separator 10.7 barg; downstream 6.1; valve 78 %; gas flow 4.5

Problem

Large pressure differential indicates downstream restriction.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses differential above 3 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when differential below 1 barg.

Expected Result

differential below 1 barg

Scenario 7: High liquid level with pressure increase

Field Condition

level 86 % from 45–65; pressure 10.4 barg; liquid flow 220 m³/h

Problem

High liquid inventory is reducing gas space and increasing pressure.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses level above 75 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when level 45–65 % and pressure 7.5–9.0.

Expected Result

level 45–65 % and pressure 7.5–9.0

Scenario 8: Gas flow reduction

Field Condition

gas flow 1.9 from 3.5 MMSm³/d; pressure 8.1 barg; liquid level 58 %

Problem

Reduced gas flow with normal pressure may indicate inlet composition, restriction, or meter error.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses flow below 2.5.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when validated flow 3.0–4.0 or approved process diagnosis.

Expected Result

validated flow 3.0–4.0 or approved process diagnosis

Scenario 9: Gas flow increase

Field Condition

gas flow 5.6 from 3.2; pressure 9.8 barg; valve 24 %

Problem

Gas surge is increasing pressure and challenging control capacity.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses flow above 4.5.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when flow 3.0–4.0 and pressure 7.5–9.0.

Expected Result

flow 3.0–4.0 and pressure 7.5–9.0

Scenario 10: Instrument disagreement

Field Condition

local pressure 8.6 barg; transmitter 10.1; test gauge 8.7; level 54 %

Problem

Pressure transmitter is reading high by 1.4 barg.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses difference above 0.5 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when transmitter within 0.3 barg.

Expected Result

transmitter within 0.3 barg

Scenario 11: Pressure-control instability

Field Condition

PID output oscillates 30–75 %; pressure 8.0–9.7; scan 5 s

Problem

Aggressive tuning, valve stiction, or noisy measurement is likely.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses output swing above 20 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when output swing below 10 % and pressure 7.5–9.0.

Expected Result

output swing below 10 % and pressure 7.5–9.0

Scenario 12: Relief-device event

Field Condition

relief lift indication active; pressure peaked 12.0 barg; normal 8.2; gas flow 5.1

Problem

A relief event indicates an overpressure scenario requiring formal response.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure above approved relief setpoint.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure stable below 9.0 barg after cause is controlled.

Expected Result

pressure stable below 9.0 barg after cause is controlled

Scenario 13: Upstream slugging

Field Condition

inlet liquid flow 40–310 m³/h; level 35–88 %; pressure 7.2–10.6

Problem

Slugging is causing rapid inventory and pressure changes.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses level change above 20 %/min.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when level change below 5 %/min.

Expected Result

level change below 5 %/min

Scenario 14: Downstream valve problem

Field Condition

separator 9.9 barg; downstream 5.8; valve command 62 %, feedback 8 %

Problem

Downstream valve is not following command and is restricting outlet.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses feedback differs by >15 points.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when feedback within 5 points and pressure normal.

Expected Result

feedback within 5 points and pressure normal

Scenario 15: Abnormal after maintenance

Field Condition

pressure 10.6 barg after control-valve work; transmitter offset 1.1 barg; valve 40 %

Problem

Incorrect range, impulse connection, or valve trim configuration is likely.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses offset above 0.5 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure indication within 0.3 barg and pressure 7.5–9.0.

Expected Result

pressure indication within 0.3 barg and pressure 7.5–9.0